

Juliana Leigh P. Yasis

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EDUCATION

B.S. Electrical Engineering | University of Oklahoma, Norman, OK *Expected Spring 2027*

Relevant Coursework: Circuits I & II, Electronics, Signals & Systems, Digital Signals & Filtering, Microprocessor System Design, Energy Conversion, Circuits & Electronics Lab

Great Plains Technology Center — Pre-Engineering Certificate, PLTW Program (2022)

TECHNICAL SKILLS

Embedded Systems: mbed OS 6, LPC1768, ESP32-WROOM-32, GPIO, ADC, UART, I2C, PWM, interrupt-driven firmware

Circuit Design: Analog PID, op-amps, BJTs, MOSFETs, 555 timers, PCB design, soldering, sensors, DC converters

Programming: C/C++, MATLAB (signal processing & simulation), HTML/CSS/JavaScript, SQL

Tools: Multisim, MATLAB/Simulink, Microsoft Office (Excel w/ macros), Google Workspace

Languages: English, Tagalog — Fluent

ENGINEERING PROJECTS

Analog Circuits Lab & PID Line Follower Robot | E-Lab ECE-3873 *April 2025 – May 2025*

- Building a fully analog line-following robot with a PID control loop using op-amp integrator/differentiator stages for the I and D terms and a 555 timer bistable circuit for motor switching logic; tuning RC time constants for stable tracking with no microcontroller.
- Built and characterized supporting circuits using op-amps (inverting/non-inverting, integrator, differentiator), BJTs, and MOSFETs; compared measured behavior against theoretical models and SPICE simulations.

LED Matrix Display System | Personal Project *Mar 2026 – Apr 2026*

- Independently designed and built a 12x8 green LED matrix display driven by an LPC1768, with multiplexed row/column scanning via PNP/NPN transistor driver circuits and mbed OS 6 firmware featuring scrolling text, animations, and hardware timer ISRs for precise refresh timing.

Non-Volatile Light Dimmer | Personal Project *2026*

- Developed a PWM-based light dimmer peripheral for the LPC1768 with user-adjustable duty cycle persisted in non-volatile memory, so brightness settings survive power cycles; implemented the NVM read/write interface in C on mbed OS 6.

LPC1768 Embedded Systems Projects | Microprocessor System Design ECE-3223 *Jan 2026 – Present*

- ADC-based digital thermometer: multi-sample averaging to reduce noise, output on a 4-digit seven-segment display.
- Reaction-time game: hardware interrupts, volatile ISR-safe state sharing, timer-based winner detection with tiebreak logic.
- Laser sound effects: piezo transducer driven by a descending-frequency PWM sweep over 500 ms, looping on button press.

PROFESSIONAL EXPERIENCE

Barista / Shift Lead | Teapioca, Norman, OK *Mar 2023 – Oct 2025*

- Trained and onboarded new staff and enforced food safety standards while managing customer transactions in a fast-paced, high-volume environment.

Water Safety Instructor | Pearland Recreation Center, Pearland, TX *Aug 2022 – Dec 2023*

- Instructed up to 40 children (ages 6 months–15 years) and 8 adults daily; adapted lesson plans to each age group and skill level while meeting Red Cross swim level requirements.

LEADERSHIP & ACTIVITIES

- Secretary & Treasurer, Lambda Delta Psi Sorority Inc. — managed records and budget, coordinated meetings, and produced newsletters and digital media.
- Treasurer, Filipino Student Association (2 yrs), and Promotions volunteer, Filipino-American Association of Lawton Fort Sill — tracked finances, managed fund distribution, and designed promotional websites, posters, and video content.

CERTIFICATIONS

Red Cross Water Safety Instructor • Lifeguard Certification • Safety Training for Swim Coaches